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Examine low cost / Free (VMWare style) options in the cloud

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***Abstract*—The purpose of this practical work is to examine low-cost / free VMware-style virtualization options available within cloud environments. The project focused on testing whether enterprise virtualization functionality could be achieved using cloud infrastructure combined with open-source virtualization technologies instead of relying entirely on a vendor-controlled virtualization platform.**

***Index Terms*—Proxmox Virtual Environment, Azure Backup, CSP comparison**

I. Introduction

The main objective of this project is to research, deploy, and evaluate cost-effective alternatives to traditional VMware infrastructure using open-source tools and native cloud service providers.

Amazon Web Services (AWS) has been providing cloud based solutions since 2006. Covering a wide range of services, from AI and databases to machine learning and serverless deployments, AWS offers a total of 175 fully-featured services across database storage, networking, compute, etc. One of the features AWS provides is the free tier which allows users to test and deploy cloud services at a very low cost for a limited time [3].

Microsoft Azure is ranked highly, alongside AWS [9]. Azure is widespread geographically across 54 regions and operating in 140 countries offering a wide variety of services including AI and machine learning, Databases, Devops, networking etc. Similar to AWS, Azure also offers free trial of 30 days period for new users.

Proxmox is not a cloud service provider itself, but is one of the leading open-source virtualization platforms used to create and manage virtual machines. Whilst VMware uses its own dedicated VMkernel hypervisor, Proxmox is based on Debian Linux and combines both KVM virtualization and LXC container technologies within the same platform [19]. In this project, Proxmox will be used within a cloud environment.

Google Cloud is renowned for its advanced data analytics and cloud infrastructure capabilities, and in this project it will be used to compare its virtualization services against both AWS and Microsoft Azure in terms of cost, deployment, performance, and management features.

To properly test and evaluate alternatives to VMware, the project will be structured into three core phases. In the first phase, AWS and Azure will be deployed and tested as a cloud-based virtualization platform. This phase will focus on creating and managing virtual machines within the cloud environment while testing core virtualization features similar to traditional VMware environments. The aim of this phase is to evaluate how AWS can provide a low-cost or free alternative to traditional on-premise virtualization platforms. A backup testing component will also be included, where backup and restore functionality will be configured and validated in both AWS and Azure to assess reliability and ease of recovery across both cloud platforms.

In the second phase, both Microsoft Azure and Google Cloud will be used to test nested virtualization by deploying Proxmox VE inside an Azure virtual machine. Unlike the first phase which focused on native cloud virtualization services, this phase will focus on recreating a more traditional VMware-style virtualization environment within the cloud. The purpose of this phase is to examine whether a low-cost cloud virtual machine can host a fully virtualized platform capable of running multiple virtual machines inside it. The project will carry out tests to identify whether a virtual machine created within a cloud-hosted Proxmox VE environment can be successfully exported and migrated back into a traditional VMware environment. This will help evaluate whether organisations can adopt open-source virtualization platforms while still maintaining the ability to return to VMware infrastructure if required.

In the third phase, the cloud platforms and virtualization solutions tested throughout the project will be compared and evaluated based on performance, usability, cost, virtualization features, ease of deployment and overall practicality.

Links to the corresponding practical work are provided through their corresponding section header.

II. AWS

Two VMs were created using Ubuntu Server, t3.micro instance and 8 GB EBS storage. The t3.micro instance type was selected as it is included within the AWS Free Tier and provides sufficient resources for this project. During the deployment of the instance, a dedicated security group was also configured to control inbound traffic, in AWS, security groups can be used similarly to firewall rules within VMWare [6].

After the deployment of both VMs, a connectivity test was completed to verify communication between VM1 and VM2 using their private IP addresses. Unlike in VMware Workstation where additional network configuration needs to be implemented, such as NAT or bridged network, both AWS VMs already had internet connection available through the default AWS VPC configuration.

SSH connectivity between the two VMs was tested using their respective IP addresses and the SSH key created during the deployement of the first instance. File transfer testing was also completed using Secure Copy Protocol (SCP) where a test file was copied successfully from VM1 to VM2 and verified after the transfer completed.

For testing purposes, an additional 1 GB Elastic Block Store (EBS) storage volume was created and attached to VM1. Unlike in VMware workstation where snapshots are usually taken of the full VM, in AWS snapshots are created of individual storage volumes [1]. This test also demonstrates how EBS storage can exist independently from the VM, meaning the volume and its snapshots can still remain available even if the VM itself is removed.

Similar to VMware workstation, in AWS a golden VM image can be created and reused to facilitate the deployment of additional VMs requiring the same operating system (OS) and configuration [2]. In AWS you can control whether the EBS storage volume is deleted automatically when the virtual machine is terminated or kept separately.

After enabling HTTP traffic in the security group, Apache was installed and tested successfully. The default Apache webpage was then replaced with a custom one summarising the work completed for this project.

III. Azure

Microsoft Azure was selected as the cloud platform for this practical work due to its ability to provide scalable virtual machine resources suitable for nested virtualization testing [13]. A Debian 12 virtual machine was deployed and configured within Azure to host the Proxmox VE hypervisor and make use of its free virtualization features.

The first step of the project involved deploying a Debian 12 virtual machine [18]. After attempting to deploy multiple virtual machines with different operating systems versions for nested virtualization, Debian 12 was ultimately selected. Different versions of Ubuntu encountered compatibility and dependency issues during the Proxmox installation process.

The Azure virtual machine was configured with Standard D4s v3 virtual machine size, 4 virtual CPUs, 16 GB RAM, Standard SSD storage, Public SSH access and Standard security configuration [15].

During the installation process several troubleshooting steps were required, particularly relating to dependency compatibility, SSL certificate generation and bridge networking configuration. The successful deployment of Proxmox VE demonstrated that enterprise-style virtualization platforms can run reliably inside cloud-hosted virtual machines.

IV. Proxmox

Nested virtualization refers to running virtual machines inside another virtualized environment and one of the main objectives of this practical work was testing nested virtualization functionality. A nested Ubuntu Server virtual machine was successfully deployed within Proxmox VE [20]. This test confirmed that Microsoft Azure could support a complete virtualization environment capable of hosting additional guest virtual machines.

Initially, two files named ”rollback.txt” and ”snapshot.txt” were created within the guest virtual machine and a snapshot of the virtual machine's state was then created using Proxmox VE. Both files were later deleted and checked to ensure they were no longer present. Next the virtual machine was restored to its previous state where both ”rollback.txt” and ”snapshot.txt” reappeared, confirming that the snapshot and recovery functionality functioned correctly [17]. The above test confirmed that Proxmox VE can preserve and restore virtual machine states, similar to enterprise virtualization platforms such as VMWare.

A full clone of the nested Ubuntu virtual machine was created successfully and powered on without issues [21]. To examine resource allocation, additional CPU and RAM resources were assigned to the cloned virtual machine. After rebooting, the guest operating system (OS) detected the updated hardware resources successfully without any issues.

In order to assess compatibility between Proxmox VE and VMware Workstation Pro the nested Ubuntu virtual machine disk was exported from Proxmox VE and converted into VMDK format to be imported into VMware Workstation Pro. The imported disk booted successfully, confirming that virtual machines created in Proxmox VE can be moved between different virtualization platforms using standard virtual disk formats. From a vendor lock-in perspective this proved that workloads on proxmox were not restricted to a single virtualization platform and could be migrated when required.

Proxmox VE provided great flexibility for virtual machine management, snapshots, cloning and migration testing whilst reducing licensing costs in comparison to competing platforms. [19]

V. Google

A further objective was to determine whether a virtual machine (VM) from the Azure-hosted Proxmox server could be imported into the Google Cloud Proxmox environment testing VM migration between cloud providers and compare the deployment experience across both platforms.

For testing purposes two virtual machines were created. The first VM was created through the Google Cloud GUI, whilst the deployment of the VM was successful, it was not possible to identify a clear option for enabling nested virtualization during the creation process. A second VM was then deployed using Google Cloud Shell and gcloud commands which forced nested virtualization to be enabled explicitly during deployment [10].

Unlike the Azure deployment, Proxmox certificate generation initially failed due to the long hostname automatically assigned by Google Cloud. To fix this, the hostname was changed to a shorter one and the certificates were regenerated. Once the certificates had been regenerated, a firewall rule was created to allow inbound access on TCP port 8006 allowing access to the Proxmox web management interface.

The virtual machine's disk was first exported from the Azure Proxmox environment in VMDK format and then transferred between both Proxmox servers using WinSCP before being imported into Google Cloud Proxmox deployment. Whilst the virtual machine booted successfully after the migration, it was unable to obtain an IP address automatically. Therefore a static address was assigned by editing the Netplan configuration file.

From a deployment perspective, Azure provided a smoother overall experience. Although some initial testing and research was required to identify a suitable operating system for hosting Proxmox VE, the overall deployment and configuration process within Azure required less troubleshooting than Google Cloud.

Google Cloud initially appeared easier due to the availability of Google Cloud Shell and command-line deployment options. However, several additional issues were encountered during deployment including nested virtualization could not be enabled easily through the graphical interface.

From a cost perspective and based on estimates obtained from the official pricing calculators [11], [12], the Azure Standard D4s v3 configuration was estimated at approximately $174.64 per month, whilst the Google Cloud n2-standard-4 configuration was estimated at approximately $190.12 per month. Although both platforms provided comparable resources, the Google Cloud deployment was approximately 9% more expensive based on the selected configurations and regions [11].

VI. Azure - Backups

An Ubuntu 24.04 virtual machine named ProdVM was created in Azure. Azure Backup was enabled during setup and recovery points were generated automatically. Once deployed, Apache was installed and configured to host the website developed during the AWS phase of the project.

To represent a small production environment, a directory named /companydata was created and populated with sample business data including customer-records.csv, orders.csv and employees.csv.

A Recovery Services Vault was automatically created as part of the backup configuration and recovery points were generated for the virtual machine [14], [16]. Backups were generated at approximately four-hour intervals. Having multiple recovery points meant that different restore points could be selected depending on when the failure occurred.

Before testing the recovery process, a Storage Account was created as Azure Backup required a staging location during the restore operation. As a precaution, an initial restore attempt was carried out while the production VM remained online. The aim was to confirm that a recovery could be completed before removing the original VM.

The operation could not be completed because the Azure for Students subscription had reached its available CPU core allocation, preventing an additional virtual machine from being created alongside the existing workload.

The original virtual machine was removed and the recovery test continued using a complete VM loss scenario. From the Recovery Services Vault, the most recent recovery point was selected and the restore operation was started.

The recovered VM was created without a public IP address. A new public IP was therefore created and attached to the network interface. Although the public IP address had been assigned, SSH access was still unavailable. The Network Security Group did not contain a rule allowing inbound SSH traffic, therefore a new rule was created to allow TCP port 22.

Apache was running and the website could be reached through the new public IP address. The contents of the /companydata directory were also present, including the customer, order and employee records created before the backup.

The production virtual machine was removed at 22:15 to simulate a failure. Access to the recovered virtual machine was restored at 22:35. At this point SSH connectivity was working, Apache was running and the business data files were available.

Some additional configuration was required after the restore. A new public IP address had to be assigned and an SSH rule was added to allow remote access to the virtual machine. Recovery points, backup configuration and restore operations were all managed through the Azure Portal. No separate backup server or dedicated storage platform was required during testing.

VII. AWS - Backups

The backup [8] configuration process was different on each platform. In AWS, a Backup Vault was created first [5], followed by a Backup Plan [4] and a Resource Assignment before the EC2 instance could be protected. An on-demand backup was then used to generate the first recovery point [7].

Azure required less manual configuration. Backup protection was enabled while the virtual machine was being deployed, after which Azure automatically created the Recovery Services Vault and applied the backup policy. Recovery points were generated without any further setup.

Both platforms successfully created recovery points that could be used to restore the virtual machine. From a deployment perspective, Azure was quicker to configure because most of the backup settings were applied automatically. AWS required a few extra configuration steps, although the separate Backup Vault and Backup Plan made it easier to organise backup policies and protected resources.

VIII. Conclusion

This practical work successfully demonstrated how Microsoft Azure, AWS and Google Cloud can be used as low-cost cloud platforms for enterprise virtualization and disaster recovery similar to a VMware environment. The practical work also demonstrated some of the differences between traditional VMware virtualization and cloud-based virtualization across the three cloud providers.

The aim of using nested virtualization in this project was to test whether an enterprise virtualization environment could be deployed and managed within a low-cost cloud platform using open-source technologies. Microsoft Azure and Google Cloud were used as the cloud infrastructure providers while Proxmox VE was deployed as the main virtualization platform responsible for creating and managing guest virtual machines. By deploying Proxmox VE inside Azure and Google Cloud, the project demonstrated that a private cloud style environment could be created and managed without relying entirely on proprietary virtualization platforms.

The comparison between Azure and Google Cloud highlighted that both platforms are capable of hosting a nested Proxmox environment, however, Azure provided a more straightforward deployment experience and was slightly cheaper. The migration test also confirmed that a VM could be moved successfully from the Azure Proxmox environment to the Google Cloud Proxmox environment. This shows that workloads do not necessarily have to remain with the same provider and can be transferred if requirements change in the future.

Compared to an on-premises environment, the recovery process was relatively straightforward. Recovery points, backup management and restore operations were all available through the Azure Portal and the AWS Management Console without requiring separate backup servers or storage systems. Although AWS Backup required additional configuration through a Backup Vault, Backup Plan and Resource Assignment, both AWS and Azure successfully restored the virtual machine, Apache web server and business data during disaster recovery testing.

With many organisations becoming increasingly concerned about vendor lock-in, especially when depending on a single cloud provider, using nested virtualization can provide an alternative approach by allowing organisations to deploy and manage their own private cloud style virtualization environment within low-cost cloud infrastructure. The successful migration between Azure and Google Cloud, together with the disaster recovery testing completed in Azure and AWS, showed that workloads and recovery strategies do not necessarily have to depend on a single cloud platform.

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